

IoT Smart Wheelchair

Arm-Gesture Control and Fall Alerts

Kubanja Elijah Eldred

24/BSE/BU/R/0012

Research Methods in Computing

Sources: WHO [1]; Arduino [2]; InvenSense [3].



Concept illustration | September
2026

Mobility and safety gaps

Study concerns—not measured local findings.

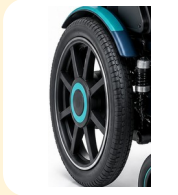
1 Limited control

Limited hand strength or dexterity can make control difficult.



2 Physical effort

Manual propulsion can be tiring and limit independence.



3 Delayed help

After a suspected fall, help may not arrive promptly.



4 Missing location

Caregivers may not know where support is needed.



Need: intuitive movement control + prompt caregiver alerts.

Four measurable objectives

Eight-week study • Planned work, not completed results.

WEEK 1

Document 10 user and caregiver requirements.

WEEK 3

Build the gesture-control model; test 100 movement commands.

WEEK 5

Integrate caregiver alerts; safely simulate 20 fall-candidate events.

WEEK 8

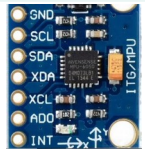
Evaluate accuracy, false alarms and responses to 10 radio-link failures.

Prototype: components and roles

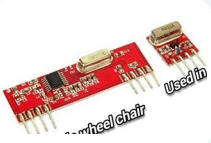
Tabletop model only—not passenger-ready.



Arduino Uno ×2
Control both units



MPU6050
Sense arm motion



433 MHz RF pair
Send commands



GPS module
Get coordinates



SIM800L GSM
Send caregiver SMS



L298N / L293D
Drive model motors



Motors + wheels
Move the model



SG90 servos*
Optional steering



Local buzzer
Sound an alert



Li-ion supply
Power the model

Key references

The same reference numbers are used in the report.

[1] WHO (2023)

Wheelchair provision guidelines.

[3] InvenSense (2013)

MPU-6000 / MPU-6050 product specification.

[5] STMicroelectronics (2023)

L298 dual full-bridge driver datasheet.

[2] Arduino (n.d.)

UNO R3 official documentation.

[4] SIMCom (2013)

SIM800L hardware design, version 1.00.

[6] Bagalà et al. (2012)

Fall-detection evaluation. PLOS ONE, 7(5), e37062.

How the proposed system works

Arm controller → wheelchair model → caregiver

1 ARM UNIT



MPU6050 + Uno
Arm gestures

➡
RF

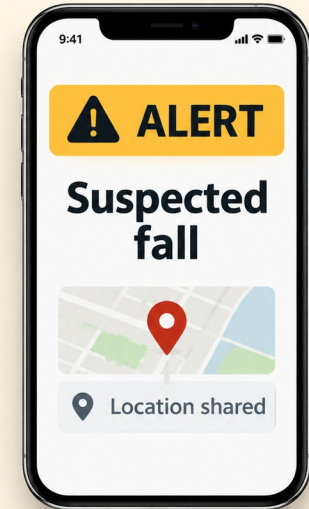
2 CHAIR MODEL



Control + drive
+ protected power

➡
GSM

3 CAREGIVER



Caregiver phone
SMS + location

Suspected fall → local alarm + SMS. *When location is available.